

Abdullah Hamdi (*Computer Vision*)

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Research Interests

3D Generative Models, 4D Understanding, Medical Image Analysis, Efficient 3D Rendering.

Academic Positions & Education

Postdoctoral Researcher, University of Oxford 2023-now

- Research focus: 3D generative AI, 4D understanding, and 3D medical imaging at VGG lab under Prof. Andrea Vedaldi

Research Fellow, Kellogg College, University of Oxford 2024-2025

- Mentoring students as part of the prestigious fellowship.

Research Intern, Technical University of Munich 2024-2025

- Led a five-month project on generating NeRFs with Prof. Matthias Niessner.

Ph.D, KAUST (GPA: 3.84/4) 2018-2023

- Computer Vision (Robustness and 3D deep learning), Electrical and Computer Engineering major under Prof Bernard Ghanem,
- Thesis: Towards Designing Robust Deep Learning Models for 3D Understanding

MS, KAUST (GPA: 4.0/4) 2016-2018

- Computer Vision, Electrical Engineering major under Prof Bernard Ghanem.
- Thesis: "Cascading Generative Adversarial Networks for Targeted Imagination"

Exchange, Texas A&M (GPA: 4.0/4) 2014-2014

- Earned 14 Credit hours in Texas A&M University, College Station TX.

BS, KFUPM (GPA: 3.97/4) 2011-2016

- Electrical Engineering in KFUPM, Saudi Arabia

Experience

Professional Service & Teaching

- **2024-now:** Junior Research Fellow at [Kellogg College](#) (Oxford), mentoring students
- **2023-2024:** Lead organizer of the "3DMV" workshop at CVPR 2024 & CVPR 2023, opening talks & facilitating discussions on multi-view supervision in 3D.
- **2024:** lectured at the Fihm.ai Summer School for over 3,800 attendees. Delivered lectures on "Intro to Deep Learning" and "3D Graphics & AI Applications."
- **2023-now:** Leading research in 3D MRI reconstruction (X-Diffusion) and 3D medical image segmentation (2 publications)
- **2023-now:** Leading research in Co4D, collecting dynamic multi-view object dataset
- **2023:** Leading a research project on 3D Gaussian Splatting at Oxford (CVPR paper)

- **2022:** Leading a research project on generating NeRFs from a few images at TUM.
- **2021:** Leading a research project on multi-view for 3D understanding (two papers).
- **2019-now:** Reviewer for 58 papers at CVPR'22, ICCV'21, NeurIPS'21, ICLR'21, ECCV'20, CVPR'20 and ICCV'19 on topics like adversarial attacks and 3D.
- **2020:** Giving [Introduction to Deep Learning Workshop](#) to +600 attendees.
- **2019:** Teaching [Lecture on GANs](#), EE354 (Intro to Computer Vision), KAUST
- **2019:** TA for AMCS 211 (Numerical Optimization MS course), KAUST

Leadership & Outreach

- **2018-now:** Founder and president of [Fihm.ai](#) ([biggest Arabic online AI platform](#)). Published 83 articles, 37 technical lessons, 32 videos, and +800K reads/views. Organized 4 summer schools (2020-2024) with attendees between 273 and 3,800.

Student Mentoring

- **Emmanuelle Bourigault** (Oxford PhD Student): Mentored on 3D medical imaging (X-Diffusion, ICCVW 2025).
- **Jan Held** (UofLiège PhD Student): Mentored on 3D rendering (Convex Splatting, CVPR 2025 Highlight).
- **Jinjie Mai** (KAUST MS Student): Mentored on 3D localization (EgoLoc, ICCV 2023 Oral) & generation (Magic123, ICLR 2024).
- **Sara Rojas** (KAUST PhD Student): Mentored on 3D robustness (AdvPC, ECCV 2020) and NeRFs.

Honors & Awards

List of Featured Publications ([Google Scholar](#))

* denotes equal contribution

Postdoc papers:

- Jiayuan Zhu, [Abdullah Hamdi](#), Yunli Qi, Yueming Jin, Junde Wu, “Medical SAM 2: Segment Medical Images as Video via Segment Anything Model 2”, arXiv preprint [Under submission to MedIA 2025]
- Emmanuelle Bourigault*, [Abdullah Hamdi](#)*, Amir Jamaludin, “X-Diffusion: Generating Detailed 3D MRI Volumes From a Single Image Using Cross-Sectional Diffusion Models”, ICCVW 2025 (ICCV Workshop on Generative AI for Biomedical Image Analysis) [[oral](#)]
- Emmanuelle Bourigault, Amir Jamaludin, [Abdullah Hamdi](#), “UKBOB: One Billion MRI Labeled Masks for Generalizable 3D Medical Image Segmentation”, ICCV 2025
- Bingchen Gong, Diego Gomez, [Abdullah Hamdi](#), Abdelrahman Eldesokey, Ahmed Abdelreheem, Peter Wonka, Maks Ovsjanikov, “ZeroKey: Point-Level Reasoning and Zero-Shot 3D Keypoint Detection from Large Language Models”, ICCV 2025
- Wenxuan Zhu, Bing Li, Cheng Zheng, Jinjie Mai, Jun Chen, Letian Jiang, [Abdullah Hamdi](#), Sara Rojas Martinez, Chia-Wen Lin, Mohamed Elhoseiny, Bernard Ghanem, “4D-Bench: Benchmarking Multi-modal Large Language Models for 4D Object Understanding”, ICCV 2025
- Jan Held*, Renaud Vandeghen*, [Abdullah Hamdi](#)*, Anthony Cioppa, Adrien Deliege, Silvio Giancola, Andrea Vedaldi, Bernard Ghanem, Marc Van Droogenbroeck, “3D Convex Splatting: Radiance Field Rendering with 3D Smooth Convexes”, CVPR 2025 [[Highlight](#)]
- Lorenza Prospero, [Abdullah Hamdi](#), Joao F. Henriques, Christian Rupprecht, “GST: Precise 3D Human Body from a Single Image with Gaussian Splatting Transformers”, CVPRW 2025 (CVSports)
- [Abdullah Hamdi](#), Luke Melas-Kyriazi, Jinjie Mai, Guocheng Qian, Ruoshi Liu, Carl Vondrick, Bernard Ghanem, Andrea Vedaldi, “GES: Generalized Exponential Splatting for Efficient Radiance Field Rendering”, CVPR 2024.

PhD & earlier papers:

- [Abdullah Hamdi](#), Faisal AlZahrani, Silvio Giancola, Bernard Ghanem, “MVTN: Learning Multi-View Transformations for 3D Understanding”, IJCV 2024 (journal extension).

- Jinjie Mai, Wenxuan Zhu, Sara Rojas, Jesus Zarzar, [Abdullah Hamdi](#), Guocheng Qian, Bing Li, Silvio Giancola, Bernard Ghanem, "TrackNeRF: Bundle Adjusting NeRF from Sparse and Noisy Views via Feature Tracks", ECCV 2024
- Guocheng Qian*, [Abdullah Hamdi*](#), Xingdi Zhang*, Bernard Ghanem, "Pix4Point: Image Pretrained Standard Transformers for 3D Point Cloud Understanding", 3DV 2024
- Guocheng Qian, Jinjie Mai, [Abdullah Hamdi](#), Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, Bernard Ghanem, "Magic123: One Image to High-Quality 3D Object Generation Using Both 2D and 3D Diffusion Priors", ICLR 2024.
- Jinjie Mai, [Abdullah Hamdi](#), Silvio Giancola, Chen Zhao, Bernard Ghanem, "EgoLoc: Revisiting 3D Object Localization from Egocentric Videos with Visual Queries", ICCV 2023 [[oral](#)].
- [Abdullah Hamdi](#), Silvio Giancola, Bernard Ghanem, "Voint Cloud: Multi-View Point Cloud Representation for 3D Understanding", ICLR 2023.
- [Abdullah Hamdi](#), Silvio Giancola, Bernard Ghanem, "MVTN: Multi-View Transformation Network for 3D Shape Recognition", ICCV 2021.
- [Abdullah Hamdi](#), Sara Rojas, Ali Thabet, Bernard Ghanem, "AdvPC: Transferable Adversarial Perturbations on 3D Point Clouds", ECCV 2020.
- [Abdullah Hamdi](#), Matthias Müller, Bernard Ghanem, "SADA: Semantic Adversarial Diagnostic Attacks for Autonomous Applications", AAAI 2020 [[spotlight](#)].
- [Abdullah Hamdi](#), Bernard Ghanem, "Towards Analyzing Semantic Robustness of Deep Neural Networks", [ECCV 2020 Workshop Proceedings](#) [[Best paper award](#)]

List of Notable Awards

- **2024:** Elected for the prestigious Junior Research Fellowship, Kellogg College, University of Oxford.
- **2023:** Best Paper Award, NeurIPS 2023 Workshop on Zero-Shot Learning
- **2022:** Winner of "Ibn Rushd Postdoc Fellowship Award", KAUST global initiative.
- **2021 & 2020:** Winner of [CEMSE Student Research Excellence Award](#) at KAUST.
- **2020:** Winner of the [NEOM AI Challenge](#), Entertainment track (AI-Sports team).
- **2020:** NVIDIA Best Paper Award at [ECCV 2020 AROW Workshop](#).
- **2014:** The first-place winner in [Nassir Bin Hamad International](#) Youth Creativity Award in Science for invention in solar dust cleaning. US Patent: US9899957B2

List of Registered US Patents

- [Abdullah Hamdi](#), "[Smart dust-cleaner and cooler for solar PV panels](#)", Granted in 2018.